

Assignment 1

Introduction to UNIX Lab (NCSC102)

Instructions:

- Once completed, show your code to the respective TAs.
 - You may ask the TAs for help if you're encountering syntax errors or having trouble understanding the questions.
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Part 1: Navigation and Directory Management

1. Use the **pwd** command to display your current directory.
2. Use the **ls** command to list the contents of your home directory.
3. Create a new directory named **NCSC101_Lab** using the **mkdir** command.
4. Navigate into the **NCSC101_Lab** directory you just created using the **cd** command.
5. Navigate back to your home directory using the **cd** command.
6. Navigate back into the **NCSC101_Lab** directory.
7. From inside **NCSC101_Lab**, create a subdirectory named **scripts**.
8. Navigate up one level from the **scripts** directory to return to **NCSC101_Lab**.

Part 2: File Operations and Permissions

1. Inside your **NCSC101_Lab** directory, create an empty file named **my_notes.txt**.
2. Add the text "This is the first line." using **echo** cmd into **my_notes.txt**.
3. Append the text "This is the last line." to **my_notes.txt** without deleting the existing content.
4. View the full contents of **my_notes.txt** using the **cat** command.
5. Copy **my_notes.txt** to a new file named **my_notes_backup.txt** using the **cp** command.
6. Rename **my_notes.txt** to **lab_report.txt** using the **mv** command.
7. Create a new directory named **backups**.

8. Move the **my_notes_backup.txt** file into the **backups** directory.
9. Use **ls -l** to view the detailed listing of **lab_report.txt**.
10. Using the symbolic method, add execute permission for the user (owner) to the **lab_report.txt** file. Verify the change with **ls -l**.
11. Using the symbolic method, remove the write permission for the group on the **lab_report.txt** file. Verify the change with **ls -l**.
12. Using the octal (numeric) method, set the permissions for **lab_report.txt** to be readable and writable by the owner, readable by the group, and have no permissions for others (**rw-r-----**). What is the command? Verify with **ls -l**.

Part 3: Text Processing and Advanced Commands

1. Use the **ls -l** command and redirect its standard output to a file named **directory_listing.txt**.
2. Run the command **ls /nonexistent_directory** and redirect only the standard error to a file named **errors.log**.
3. Use the **ls** command and pipe its output to the **grep** command to find only files and directories with "lab" in their name.
4. Use the **grep** command to search for the word "first" in your **lab_report.txt** file.
5. Count the number of lines in **directory_listing.txt** using the **wc** command.
6. Display only the first 5 lines of the **/etc/passwd** file using the **head** command.
7. Display only the last 5 lines of the **/etc/passwd** file using the **tail** command.
8. Construct a single command that lists all items in your current directory, sorts them in reverse alphabetical order, and displays the result on the screen.